Reconfigurable Modular Antenna System With Null Steering and Circular Polarization

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***Abstract*— The design and analysis of a modular antenna system that can achieve both multidimensional null steering and dual-sense circular polarization (CP) is presented. The system is built using compact elementary units, each comprising a wideband tunable phase shifter and a printed dipole, allowing a modular assembly for various operational modes. The tunable phase shifter, designed with a broadside coupling structure and varactor-tuned stubs, provides a 145° tuning range with ±2° phase imbalance and less than 1 dB insertion loss across the band 5–7 GHz. Reconfiguration is achieved by tuning the phase shifters, enabling both 1-D and 2-D null steering as well as dual-sense CP. In its 1-D configuration, the system achieves a null steering range of ±40° from the broadside with more than 15 dB null depth. The 2-D configuration extends this steering capability to two planes. A different configuration of modules enables left-hand and right-hand CPs with high purity, maintaining an axial ratio below 2.3 dB across the operation band. Experimental results align closely with simulations, underscoring the potential of compact, tunable phase shifters for versatile reconfigurable antenna systems.**

***Index Terms*— Antenna, circular polarization (CP), null steering, phase shifters.**

I. Introduction

RECONFIGURABLE antennas represent a key advancement in modern wireless communications systems, offering adaptability to varying environmental conditions and communications requirements. These antennas can dynamically reconfigure their properties, such as frequency [1], [2], [3], [4], polarization [5], [6], [7], radiation pattern [8], [9], or perform compound reconfiguration of several properties [10]. Several methods, such as electronic tunable elements, optical switches, and phase shifters, have been employed in reconfigurable antenna designs to enable this adaptability, each with its advantages and limitations [11], [12], [13], [14], [15], [16].

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A tunable phase shifter is a critical component in modern reconfigurable antennas and phased arrays, enabling both beam scanning and reconfigurable beam shaping. To achieve a broad scanning range, wide operating bandwidth, and precise radiation patterns, a high-performance phase shifter with minimal in-band phase and magnitude imbalances is required.

Over the past few decades, various tunable phase shifters have been explored using different technologies. In [17], a continuously tunable filtering phase shifter employing external coupling networks and internal resonances was presented. A similar approach with external coupling networks and resonant structures was reported in [18]. The design in [19] utilized a power dividing circuit and two 3-dB hybrids, while [20] adopted a resonator-based approach. In [21], a 3-dB hybrid and reflection loads were used, and Ebrahimi et al. [22] implemented composite inductive-capacitive loaded slow wave transmission lines. However, these structures exhibit a relatively high phase and magnitude imbalance across the operating bandwidth. Moreover, all these designs rely on varactor diodes for phase tunability, where the number of diodes and independent biasing networks impact both the compactness and simplicity of the design.

One significant application of reconfigurable antennas is null steering, a technique used to suppress interference or enhance signal reception by adjusting the antenna's radiation pattern to minimize radiation in specific directions. This technique is also utilized in indoor angle-of-arrival estimation systems [23]. By electronically steering nulls toward interfering sources while maintaining signal strength in desired directions, null steering effectively improves communications reliability and spectral efficiency in crowded wireless environments [23], [24], [25], [26]. Various research efforts have explored null steering techniques, including adaptive arrays [27], where optimization techniques have been developed to find the best amplitude/phase excitation and optimal placement of physically irregular array elements for directing nulls toward jammers. For example, a microstrip patch antenna with shorting pins for null steering and radiation pattern control was presented in [28]. This antenna, loaded with four integrated varactors, enabled continuous adjustment of its radiation pattern while operating at a consistent frequency, achieving pattern tilting angles from 19° to 54° with a null depth of around 10.9 dB. A similar concept was applied to a multiport antenna in [29], which introduced shorting pins within the antenna cavity, resulting in an

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operational bandwidth of 1%. Another study [24] introduced a null steering antenna devised through characteristic mode analysis, capitalizing on an asymmetric excitation strategy with phase disparities between two capacitively coupled elements above a conductive plane. This design achieved a null-scanning capability of up to 32°, with a null depth surpassing 18 dB at a fixed operating frequency of 2.45 GHz. Additionally, a compact microstrip antenna equipped with 12 integrated varactors was introduced in [25], allowing continuous null-scanning capability at an angle of up to 37°, with a 3.3% frequency range of operation and a null depth of around 10 dB.

Null steering antenna systems are characterized by critical performance metrics, including null steering scanning range, null depth, and operating bandwidth. Additional factors common to all antenna systems, such as compactness, cost-effectiveness, and adaptability to varying environments, also significantly influence performance. Traditionally, phased array antennas have employed expensive integrated chips for amplitude and phase weighting to achieve null steering. While effective, this approach often results in bulky and costly systems. In contrast, alternative null steering techniques offer more compact and economical solutions, though these typically exhibit limited steering ranges, shallower null depths, and reduced flexibility compared to phased array systems. To address these limitations while maintaining cost-efficiency, incorporating tunable phase shifters into null steering systems presents a promising approach.

As communications technologies advance, there is a growing need for more versatile antenna systems capable of not only radiation control but also polarization adaptability. A modular antenna system is proposed to address this need. The system comprises separate units with a compact tunable phase shifter and a wideband antenna element. The tunable phase shifter, designed with a broadside coupling structure and varactor-tuned stubs, shows notable phase and magnitude balance across a wide frequency band compared with existing tunable phase shifters. This design is crucial for achieving null steering and circular polarization (CP), enhancing the system's versatility. By arranging these units in different configurations, the system can be customized to achieve various functionalities, including 1-D and 2-D null steering, as well as the generation of CP with reconfigurable right-hand or left-hand sense. This modular approach allows for flexible design and reconfiguration, enabling the creation of multiple system architectures from a common set of components.

The main contribution of this work is the introduction of wideband tunable phase shifters integrated into a modular antenna system designed for versatile applications. The phase shifter uses a broadside coupling structure with varactor-tuned stubs. The device achieves minimal phase and amplitude imbalance over a wide bandwidth. Furthermore, using single biasing for each phase shifter simplifies control and integration. Utilizing these tunable phase shifters, a reconfigurable modular antenna system is designed and tested. The modular antenna is capable of multidimensional null steering over wide angles. The design also supports dual

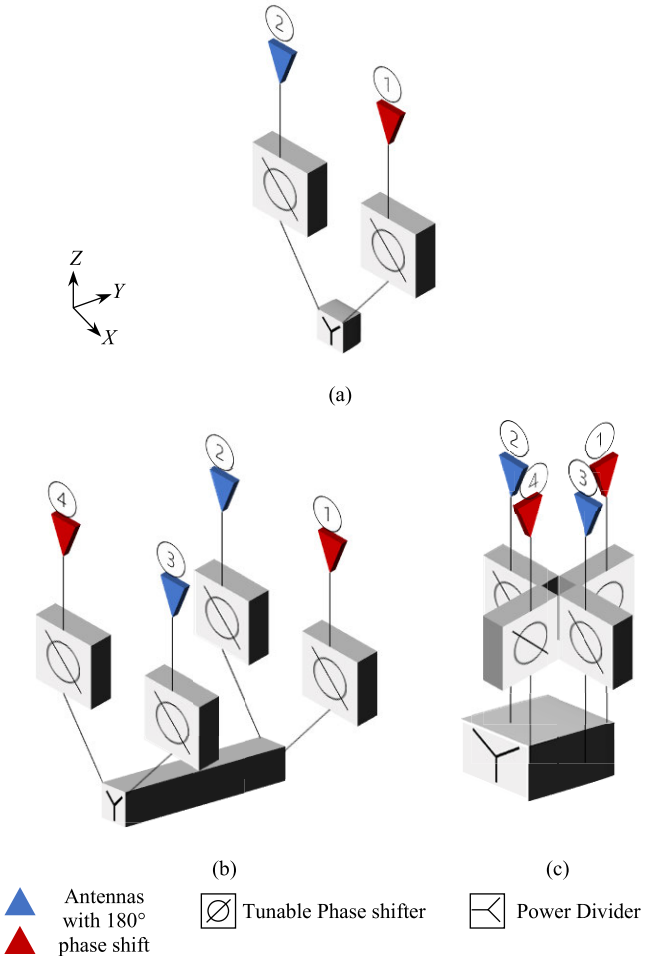


Fig. 1. Schematic of the proposed reconfigurable antenna system. (a) One-dimensional null steering system utilizing tunable phase shifters of symmetrical out-of-phase units. (b) Two-dimensional null steering system utilizing tunable phase shifters of symmetrical out-of-phase units. (c) Antenna system with LHCP or RHCP. The two-phase states are represented in red and blue colors.

CP senses (right-hand and left-hand) in a reconfigurable configuration.

This article is organized as follows: Section II details the system architecture, design concept, and operational principles of the proposed antennas. Section III discusses the design of system components, including the tunable phase shifter and the antenna design. Section IV presents the antenna systems along with the experimental results of the prototypes, with a focus on their null steering capabilities and CP generation. Finally, Section V concludes the article.

II. Systems Architecture

The proposed system is a versatile modular structure. Each unit in the system comprises a balanced tunable phase shifter paired with a wideband printed dipole, both designed to operate seamlessly within the same bandwidth (see Fig. 1). This modular approach enables flexible reconfiguration, allowing various system architectures, including 1-D and 2-D null steering and dual-sense CP generation.

In the 1-D null steering system, two units are arranged in the same plane, with the feeding network mirrored in the

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xz plane as shown in Fig. 1(a). This configuration ensures that the antennas are 180° out-of-phase, resulting in a deep null in the combined radiation pattern. By adjusting the phase of one unit while keeping the other constant, the null can be steered across a ±40° range along the axis on which the units are aligned with the null depth of more than 15 dB.

The 2-D null steering system is achieved by using a 2 × 2 array of units. The spacing for these antennas is also kept at $\lambda_o/2$, with a null steering range of ±40° in both the *xz* and *yz* planes with the null depth of more than 10 dB [Fig. 1(b)]. The four antennas are oriented to ensure that each pair of antennas along the *x*-axis or *y*-axis is out-of-phase, that is, each adjacent pair of antennas (1, 2), (3, 4), (1, 3), and (2, 4) are out of phase by 180° [Fig. 1(b)]. Without additional phase shift, this configuration forms a radiation pattern with a null in the perpendicular planes along the *x*-axis and *y*-axis, enabling a 2-D null steering. To provide a reconfigurable sense of CP, that is, either LHCP or RHCP. Specifically, four units of the proposed integrated antenna with phase shifters are utilized in the arrangement of Fig. 1(c).

III. System Components Design

A. Continuously Tunable Phase Shifter

A compact, low-cost tunable phase shifter is introduced using the broadside-coupled concept [30], as illustrated in Fig. 2. The design consists of two broadside-coupled microstrip patches positioned on the top and bottom layers of a two-substrate layer structure, where the two substrates are bonded together using a prepreg material. The two patches are connected at one of their terminals to either the input or the output feeder, while the opposite terminals are linked to a stub loaded with a varactor diode. The central metal layer is a ground plane that separates the coupled patches. A coupling slot is embedded in this ground plane to control the coupling between the top and bottom layers. The phase of the microwave signal passing through this configuration can be controlled by adjusting the coupling factor of the coupled structures, and the varactor capacitance.

To analyze the proposed structure, the conventional methodology for coupled microstrip lines [31], [32], [33] is utilized. The phase shifter is treated as a four-port backward coupler, where two output ports are terminated in open circuits, while the remaining two serve as the input and output ports [Fig. 2(a)]. This coupled structure can be analyzed using even-odd mode analysis to calculate the reflection (S_{11}) and transmission (S_{21}) coefficients [34]

$$S_{11} = \frac{\left[1 - C^2\left(1 + \sin^2\left(\beta_{ef} l\right)\right)\right]}{\left[\sqrt{1 - C^2}\cos\left(\beta_{ef} l\right) + j\sin\left(\beta_{ef} l\right)\right]^2} \tag{1}$$

and

$$S_{21} = \frac{\left[j2C\sqrt{1 - C^2}\sin\left(\beta_{ef} l\right)\right]}{\left[\sqrt{1 - C^2}\cos\left(\beta_{ef} l\right) + j\sin\left(\beta_{ef} l\right)\right]^2}. \tag{2}$$

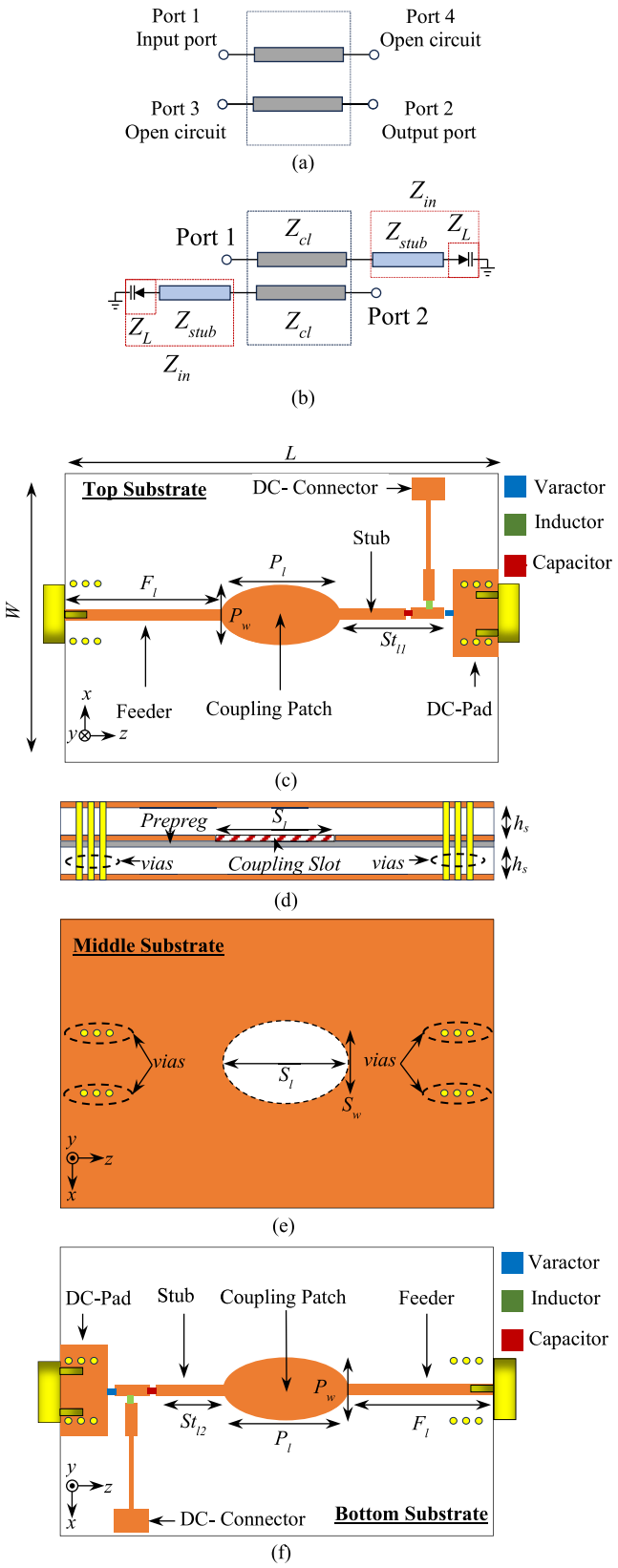


Fig. 2. Proposed tunable phase shifter. (a) Schematic of four-port backward coupler, (b) schematic of the proposed tunable phase shifter, (c) top view, (d) side view, (e) middle layer containing coupling slot, and (f) bottom view. Dimensions (mm): $L = 36$, $W = 25$, $F_l = 13.7$, $S_w = 8.8$, $S_l = 8.8$,$l_{\text{stub}} = 9.4$, $St_{l2} = 6.1$, $l = 8.8$, $P_w = 4.9$, $h = 0.508$, and *prepreg* $= 0.1$.

However, the coupled structure is connected to varactor-loaded stubs. The reflection coefficient of the stubs

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($\Gamma_{\text{loaded_stub}}$) is [see Fig. 2(b)]

$$\Gamma_{loaded_stub} = \frac{Z_{in} - Z_{cl}}{Z_{in} + Z_{cl}} \tag{3}$$

where Z_{in} represents the input impedance of the stub-loaded section [as defined in (4)], while Z_{cl} is the characteristic impedance of the coupled patches defined by the dimensions of the coupling patches and properties of the substrate. The input impedance Z_{in} is

$$Z_{in} = Z_{stub}\left[\frac{Z_L + jZ_{stub}\tan(\beta_{st}l_{stub})}{Z_{stub} + jZ_L\tan(\beta_{st}l_{stub})}\right] \tag{4}$$

where Z_{sub} is the characteristic impedance of the stubs, defined by the stub's width and properties of the substrate. The phase constant of the stub is β_{st}, while l_{stub} is the physical length of the stub.

The varactor can be emulated by its equivalent circuit that has an impedance of

$$Z_L = R_p + \frac{1}{j\omega C_i} + j\omega L_p. \tag{5}$$

R_p represents the varactor's equivalent resistance, while C_i is its variable junction capacitance, while L_p is its parasitic and terminals inductance. The reflection and transmission coefficients of the device with varactor-loaded stubs can then be calculated as

$$S_{11} = \frac{\left[1 - C^2\left(1 + \sin^2\left(\beta_{ef}l\right)\right)\right]\Gamma_{loaded_stub}}{\left[\sqrt{1-C^2}\cos\left(\beta_{ef}l\right) + j\sin\left(\beta_{ef}l\right)\right]^2} \tag{6}$$

$$S_{21} = \frac{\left[j2C\sqrt{1-C^2}\sin\left(\beta_{ef}l\right)\right]\Gamma_{loaded_stub}}{\left[\sqrt{1-C^2}\cos\left(\beta_{ef}l\right) + j\sin\left(\beta_{ef}l\right)\right]^2}. \tag{7}$$

The length and width of the loaded stubs play a significant role in tuning and determining the phase range. In the design process, emphasis was placed not only on the phase range but, importantly, on minimizing in-band phase error and insertion loss. Using (6) and (7), the relationship between stub length and the maximum phase deviation and insertion loss across different tuning ranges can be calculated as shown in Fig. 3. The stub length should thus be selected as a quarter wavelength. In this case, it functions as an impedance transformer, improving matching between the coupled structure and the varactor. Subsequently, fine-tuning is performed to adjust the phase and insertion loss of the phase shifter. Then, each of these stubs is equipped with varactor diodes at its side edges, with each varactor connected to the ground plane of the structure, which is formed by shorting the middle ground plane of the phase shifter with a top pad. The varactor diode (MAVR-000120-14110P) is from MACOM with $C_{i(min)} = 0.14$ pF, $C_{i(max)} = 1.7$ pF, $R_p = 2\ \Omega$, and negligible L_p. In this design, symmetry is crucial for maintaining the performance of the proposed phase shifter.

While the dimensions of the coupling patches and stubs are fixed, the tunability of the phase is controlled by two varactors, which are biased equally. With two varactors, a symmetric tuning response across the phase shifter is achieved. Placing them symmetrically results in the same impedance from the two sides, maintaining balanced tuning and reducing phase asymmetry. The phase shift can then be adjusted by varying a single applied voltage onto the varactors through the biasing circuit, which results in the capacitance C_i changes. To ensure suitable biasing and driving of the varactor, careful attention has been paid to designing the biasing circuit as shown in Fig. 2. A close examination of the biasing circuit of the varactor reveals that the biasing network consists of a choke inductor of 100 nH connected after the applied dc voltage. Additionally, a dc-blocking capacitor of 39 pF is inserted in the stub. The bias circuit is employed to control the tunable varactor capacitances C_i from 0.14 to 1.7 pF. The design is implemented on a Rogers RO4003C substrate with a thickness $h_s = 0.508$ mm, relative permittivity $\varepsilon_r = 3.55$, and loss tangent $tan\delta = 0.0027$. To attach the substrates, a Rogers 4450F prepreg layer with a thickness of 0.1 mm, relative permittivity $\varepsilon_r = 3.52$ and loss tangent $tan\delta = 0.004$ is used.

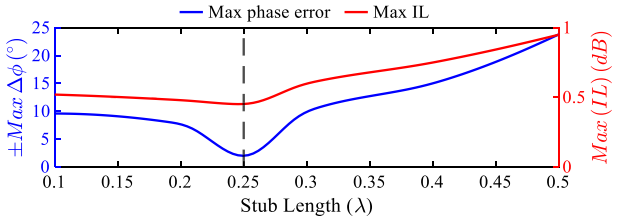


Fig. 3. Effects of stub length on in-band maximum phase error and insertion loss based on analytical calculations (6) and (7).

The analytical evaluation of the proposed design is based on (6) and (7), considering various tuning values of the capacitance C_i and the parameters of the varactor diodes. The analytically calculated S-parameters for the tunable over compact phase shifter are shown in Fig. 4(a). The phase shifter is expected to demonstrate insertion loss lower than 1 dB across the 5 to 7 GHz frequency band, indicating a wideband performance within the varactor diode tuning range of $C_i = 0.14$–1.7 pF. The return loss at the input port of the coupler exceeds 20 dB within this frequency range. Additionally, the phase shifter can achieve a phase shift tunable over a range of around 150°, as illustrated in Fig. 4(b).

For numerical validation, simulation results are carried out using the time-domain solver of Microwave Studio from computer simulation technology (CST). The simulated tunable phase range associated with the tunable varactor is 145°, and it exhibits only ±2° phase imbalance, in the worst case over the edges of the band of operation and insertion loss lower than 1 dB across the wide operation bandwidth of 5–7 GHz. The phase tuning for different capacitances is shown in Fig. 5(a) and (b).

The analytical and numerical evaluations of the tunable phase shifter show strong agreement, with both predicting a tunable phase range of approximately 145° to 150° and insertion loss below 1 dB across a wideband extending from 5 to 7 GHz. This consistency confirms the accuracy and effectiveness of the proposed design.

The phase shifter is designed as the feeding network for a reconfigurable antenna system, addressing key challenges found in conventional phase shifters. This compact design

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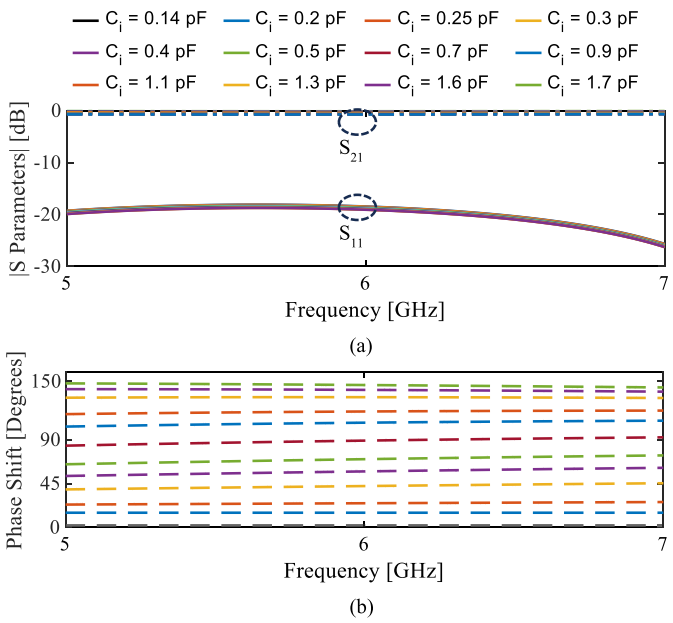


Fig. 4. Analytically computed performance of the proposed tunable phase shifter. (a) S-parameters and (b) phase responses for different varactor capacitance values.

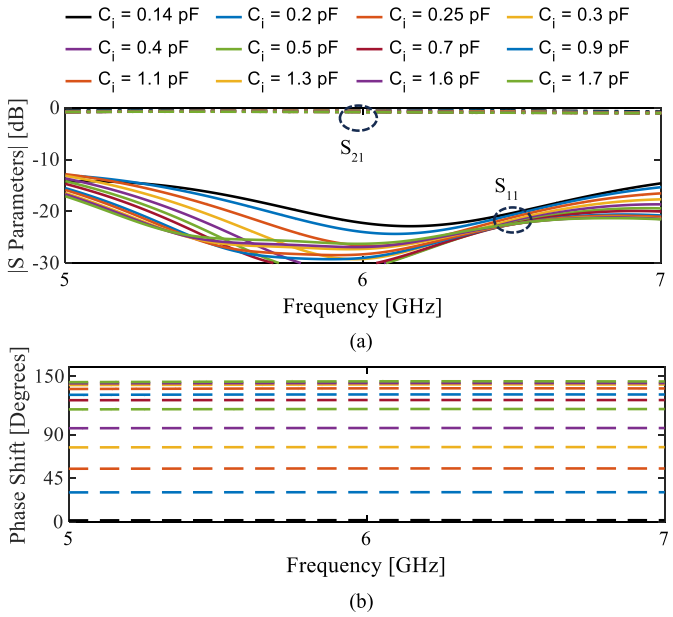


Fig. 5. Numerical validation of the performance of the proposed tunable phase shifter was conducted through simulations using CST. (a) S-parameters and (b) phase responses for different varactor capacitance values.

demonstrates superior performance in its phase and amplitude balance. Table I shows a comparison between the proposed phase shifter and previously reported phase shifters. The proposed device demonstrates significantly improved performance across most of the metrics. Operating within a 5–7 GHz bandwidth and providing a phase range of 145°, it achieves a low insertion loss of 0.8 ± 0.15 dB, outperforming [18] with 3.6 ± 0.5 dB and [19] with 4 ± 10.6 dB. Additionally, it delivers a competitive fractional bandwidth of 33.3%, whereas most

TABLE I
COMPARISON WITH PREVIOUSLY REPORTED TUNABLE PHASE SHIFTERS

Ref.	Frequency range (GHz)	Insertion loss (dB)	FBW (%)	Phase error (°)	Phase range (°)	No. Varactors	No. Bais voltages
[17]	1.07-1.36	1.1±0.3	24	±5	90	6	3
[18]	1.5-1.95	3.6±0.5	26.1	±5.5	45⁺	6	3
[19]	2-3	4±10.6	40	±8.6	104	2	2
[20]	6.1-6.9	3.2±0.35	12	±14	360⁺	20	8
[21]	2.25-2.75	1.28	20	±5.8	146	4	2
[22]	1.9-2.1^	2.1±0.4^	10	>10^	140	4	1
Proposed	5-7	0.8±0.15	33.3	±2	145	2	1

⁺ The phase range includes two cascaded units.
^ The value estimated from given statistics

other designs exhibit fractional bandwidths between 10% and 40%. Notably, the phase imbalance is only ±2°, a substantially tighter specification over comparable models like [19], [20], which show phase imbalances of ±8.6° and ±14°, respectively. Moreover, the phase shifter requires only two varactors and one bias voltage, offering simplicity and efficiency in contrast to designs in [17], [18], and [20], which use between 4 and 20 varactors and three or more bias voltages.

B. Antenna Element Design

For the proposed application, a compact endfire antenna is needed with an operation bandwidth corresponding to that of the phase shifter. Planar antennas are widely used due to their ease of fabrication, cost-effectiveness, and potential for designs with wide bandwidth and high efficiency. Various types of planar antennas have been employed in phased-array systems, including quasi-Yagi antennas [35], printed dipole, and angled printed-dipole [36]. Additionally, T-dipole fed by integrated baluns have been extensively developed for microwave wireless communications [37]. While these antennas can achieve wideband or multiband operations, their drawback lies in their larger physical size. In this work, a wideband printed dipole integrated with a balun is utilized. This antenna has a broad main beam that meets the needs of applications that require a null steered within that wide beam. Furthermore, due to the broad radiation pattern, steering the null direction has a minor effect on the main beam direction.

Advantages of this single-element geometry include compact size, broad impedance-matching bandwidth, stable radiation patterns, small gain variations over the band of operation, and high radiation efficiency. The structure of the printed dipole is shown in Fig. 6. It is patterned on both sides of a Rogers RT/Duroid 4003C substrate ($h_s = 0.508$ mm, $\varepsilon_r = 2.2$, and $\tan\delta = 0.0009$). The antenna's components include a 50 Ω microstrip-line feed, a truncated ground plane, an integrated balun, and a printed dipole. The feedline is located on the substrate's top layer, while the dipole and the ground plane are on the bottom layer. The dipole receives power from a slot line with a characteristic impedance of 116 Ω. The balun, acting as a microstrip-to-slot-line transformer, aids in achieving impedance matching across a wide frequency range. This is accomplished by adjusting the folded feed line and rectangular slot. The gaps in the slotline (g_s) and the strapline (l_7) are key design parameters for impedance matching. The

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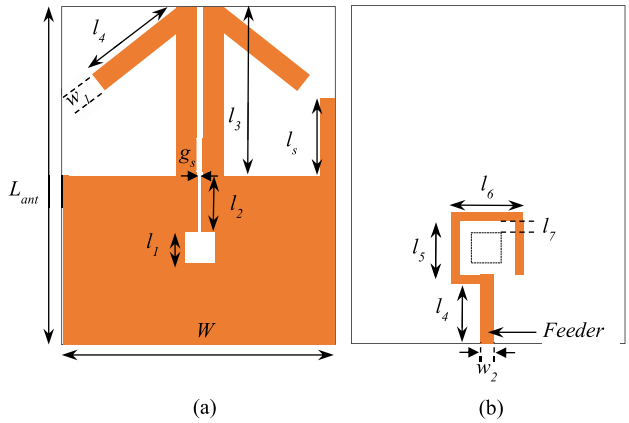


Fig. 6. Printed dipole. (a) Top view and (b) back view. Dimensions (mm): $L_{ant} = 24$, $W = 25$, $w_1 = 1.6$, $w_2 = 1.07$, $l_1 = 1.65$, $l_2 = 3.1$, $l_3 = 14.2$, $l_4 = 4.5$, $l_5 = 4$, $l_i = 4.1$, $l_7 = 0.43$, $g_s = 0.11$.

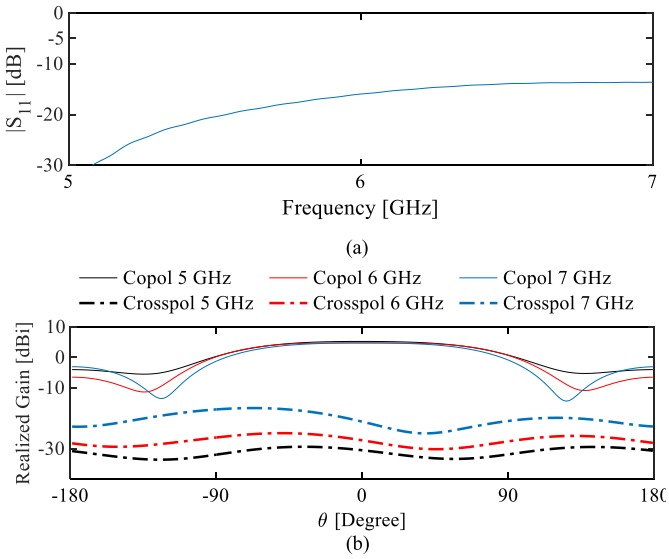


Fig. 7. Simulated performance measures of the proposed antenna. (a) Reflection coefficient and (b) radiation pattern.

printed dipole is angled at 45° to achieve a compact size and a wide E-plane pattern.

The simulated return loss for the printed dipole is lower than 10 dB for frequencies from less than 5 GHz to more than 7 GHz which is within the operation range of the tunable phase shifter [Fig. 7(a)]. The simulated radiation pattern of the antenna is shown in Fig. 7(b) for the co- and cross-polarization pattern for frequency ranges 5–7 GHz with a realized gain of around 5 dB.

IV. Systems Design and Discussion

This section presents and discusses the simulated and measured results of the three proposed systems: the 1-D null steering system, the 2-D null steering system, and the reconfigurable circularly polarized antenna system.

A. One-Dimensional Null Steering

The 1-D null steering system configuration is shown in Fig. 8. In this configuration, the system can steer a null along

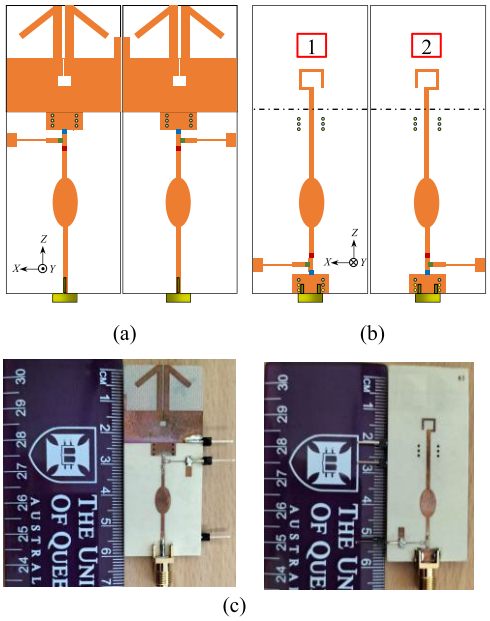


Fig. 8. One-dimensional null steering antenna. (a) Front view, (b) back view which shows the feeding network, and (c) photographs of the fabricated prototype.

the $\pm x$-axis up to an angle greater than ±40° with a null depth of more than 15 dB. This is achieved by combining the tunable phase shifter and the antenna into a unit of the reconfigurable antenna.

The unit has been designed, fabricated, and tested (Fig. 8). The reflection coefficients for single elements with different applied bias voltages of the phase shifter have been tested, and the results are shown in Fig. 9(a). Then, the system is built using an array antenna with two mirrored elements separated by $\lambda_o/2$ at 6 GHz, each connected to a tunable phase shifter. The units were connected to a (ZX10-2-183-S+) two-way splitter. The measured tuning range of the antenna operating frequency in this state is from 5 to 7 GHz (or 33.3% fractional bandwidth) as shown in Fig. 9(b). The differences between the simulated and measured results can be explained by the fact that the ideal coupler used in simulations slightly differs from the built lossy coupler.

The challenge with the antenna elements lies in the mutual coupling between the two antennas. By introducing a stub between the two antennas, the mutual coupling can be minimized. This stub has been designed according to the microstrip filter theory [17].

The simulated and measured radiation patterns at frequencies from 5 to 7 GHz for different cases of null steering at 0°, ±25°, and ±40° are shown in Fig. 10. There is a strong correlation between the simulated and measured results, indicating that a continuously steerable null scanning of wide-angle range up to ±40° is achieved with more than 15 dB null depth. A slight angular shift of less than ±3° between the simulated and measured pattern can be noticed, which is attributed to fabrication and measurement imperfections. The simulated and measured realized gains for null steering at 0°, ±25°, and ±40° across the specified frequencies show good agreement. The measured gain ranges from 4.6 to 7.8 dBi, with slightly higher values observed when steering toward 25° and 40°. The

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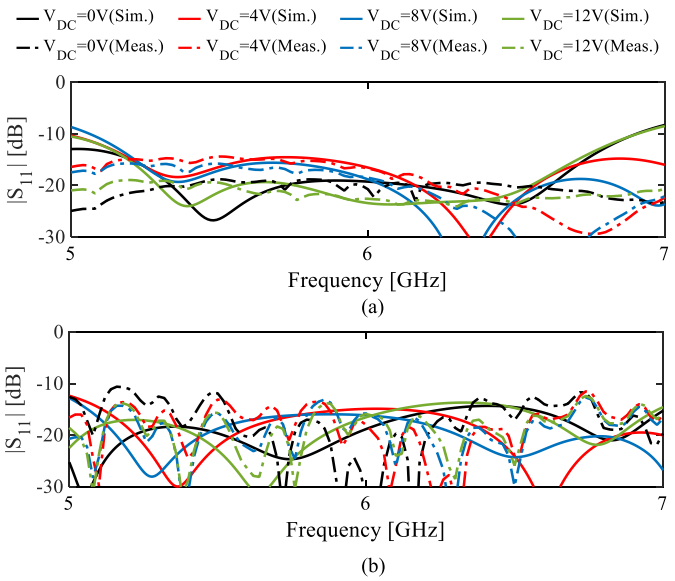


Fig. 9. One-dimensional null steering system. (a) Simulated and measured single unit reflection coefficient and (b) Simulated and measured antenna reflection coefficient.

results indicate that the measured gains are consistent with the simulated predictions, demonstrating reliable performance across the different steering angles [Fig. 10(d)].

The selected antenna exhibits a broad beam radiation pattern, which meets the requirements for the proposed null steering application. Although the 1-D radiation patterns may appear to include a sidelobe, this is, in fact, a broad beam with a null created within it. Furthermore, due to the broadness of the radiation pattern, steering the null direction has only a minor effect on the main beam direction without a significant impact on its orientation.

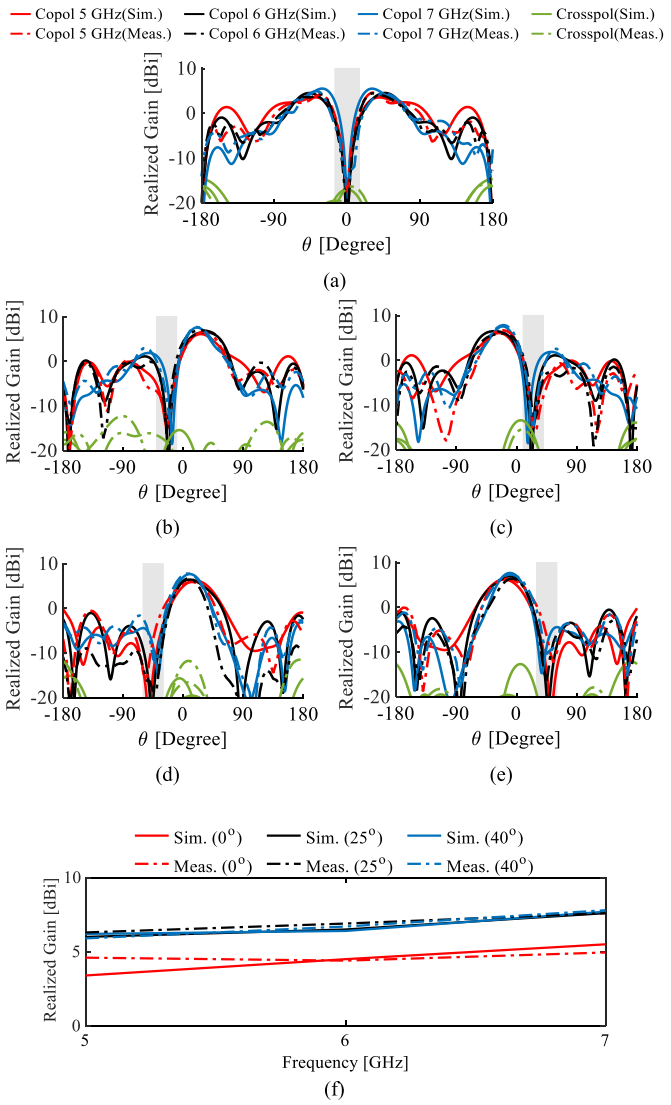


Fig. 10. One-dimensional null steering gain patterns in the xz-plane. (a) Null scanning at 0°, (b) null scanning at −25°, (c) null scanning to +25°, (d) null scanning at −40°, (e) null scanning at +40°, and (f) realized gain.

B. Two-Dimensional Null Steering

Arranging four units in a 2 × 2 antenna array enables the system to achieve 2-D null scanning. The four antennas are configured so that each pair along the x-axis and y-axis is out of phase, creating a radiation pattern with a null in the perpendicular plane. This setup allows for 2-D null steering by individually controlling the antennas along each axis. In this 2 × 2 array, each adjacent pair of antennas is out of phase by 180°, as shown in Fig. 11. The units were connected to a (ZN4PD-5R183-S+) four-way splitter. To clearly illustrate the 2-D null steering system, the uv plotting of the radiation patterns has been chosen. The simulated and measured radiation patterns at 6 GHz for the null steering at 0° and ±40° over the x-axis and y-axis are shown in Fig. 12. The measured gain ranges from 6.1 to 8.1 dBi. The simulated and measured results show a strong correlation.

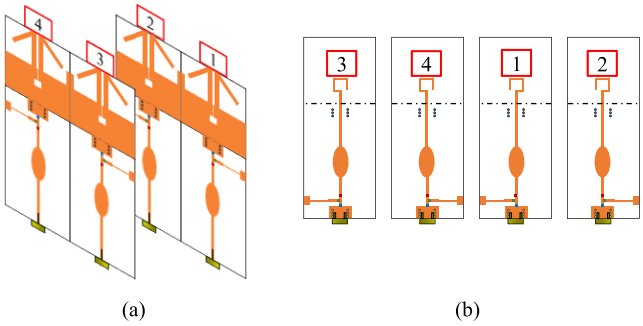


Fig. 11. Two-dimensional null steering. (a) Front view and (b) back view showing the feeding network.

C. Circularly Polarized Antenna With Reconfigurable Sense

Circularly polarized antennas are an essential requirement in satellite communications. The CP of the antenna array can be achieved by utilizing CP antenna elements or by specific feeding methods. However, an antenna array with CP may have limitations in terms of bandwidth and polarization purity.

The units are strategically positioned to form a CP feeding network, which is essential for achieving the desired polarization. Arranging the antenna in a cross-configuration, the

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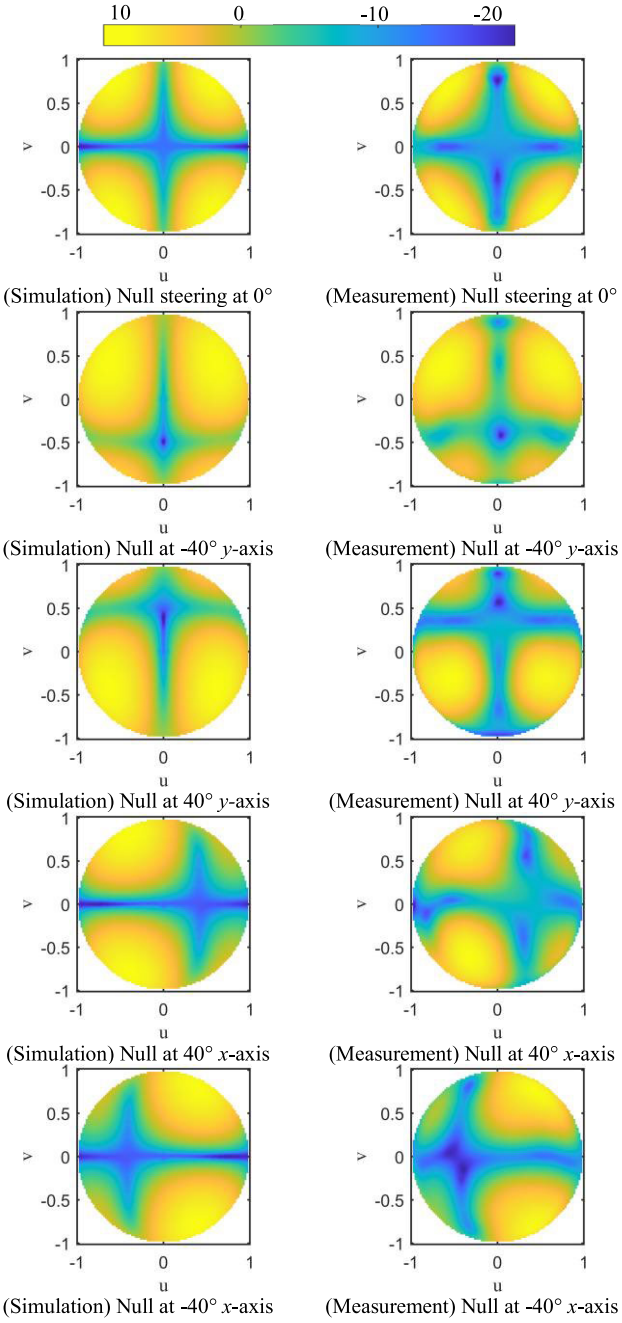


Fig. 12. Two-dimensional null steering system showing simulated (left column) and measured (right column) null steering presented as a *uv* projection.

system can generate both RHCP and LHCP, depending on the phase settings. Specifically, by adjusting the phase shifters to 90° for units (1 and 4) or (2 and 3), the system can produce RHCP or LHCP, respectively. This setup allows for different-side control over the polarization states, making it suitable for various applications where different polarizations are required. The system configuration is depicted in Fig. 13, illustrating the arrangement of the units, their respective phase shifters, and the measurement setup. The simulated and measured radiation patterns are presented in Fig. 14, highlighting the system's ability to accurately generate the desired CP. This demonstrates the effectiveness of cross-configuration in achieving the intended polarization states. The axial ratio

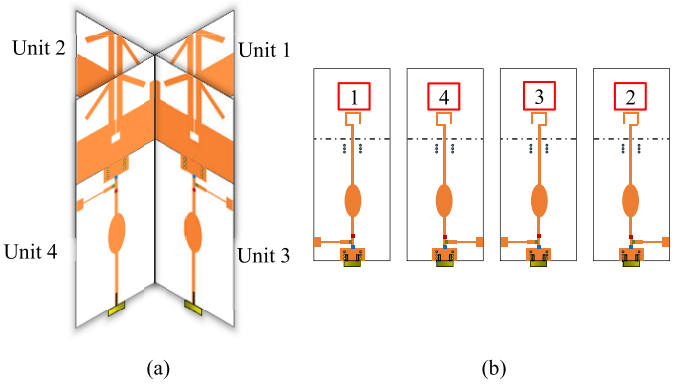


Fig. 13. Circularly polarized antenna configuration. (a) Perspective view and (b) back view showing the feeding configuration.

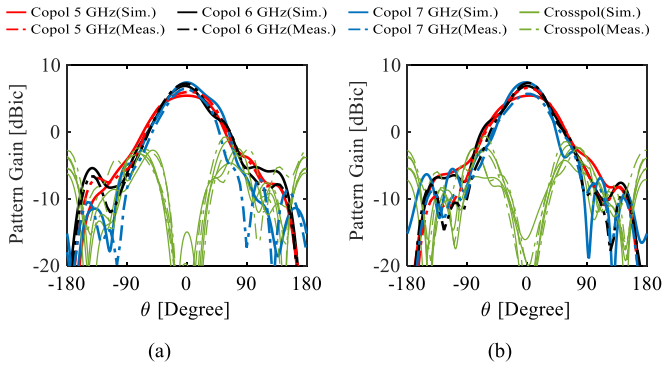


Fig. 14. Simulated and measured radiation pattern. (a) LHCP and (b) RHCP.

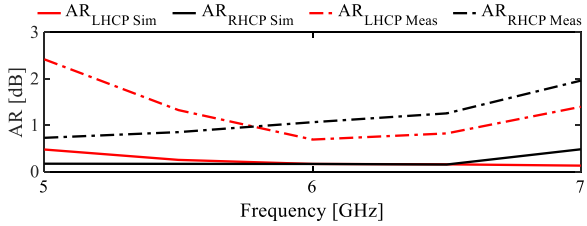


Fig. 15. Simulated and measured axial ratio for the LHCP and RHCP states of the reconfigurable antenna.

bandwidth for both the simulated and measured antenna systems is presented in Fig. 15 for the cases of LHCP and RHCP. The CP purity is high for both antenna arrays, with values below 2.3 dB in the worst case, which demonstrates the reliability of the phase shifters. The system achieves a CP over a wide frequency range from 5 to 7 GHz.

D. Comparison With Previous Works

Table II provides a comparison of the proposed null steering system with notable existing designs. The proposed network demonstrates significantly improved performance compared to others concerning the scanning capabilities and antenna performance. Starting with the operation bandwidth, the proposed system has a fractional bandwidth of 33.3% with null steering over an angle greater than ±40°, and with more than 15 dB null depth for the 1-D null steering system and more than 10 dB for the 2-D null steering system. Furthermore, it uses

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TABLE II
COMPARISON WITH PREVIOUS SINGLE-NULL STEERING ANTENNAS

Ref.	Frequency range (GHz)	Null steering angular range (°)	Null depth (dB)	Scanning modality	Gain (dBi)	Reconfigurability mechanism	Dimensional angular scanning
[23]	2.21 to 2.59 (16%)	-28° to 28°	15	Continuous	4.1 to 7.5	4 varactors	1-D
[24]	2.45	0° to 32°	18	Discrete	N/A	2 capacitive excitors	1-D
[25]	2.38 to 2.47 (3.3%)	0° to 37°	10	Continuous	4 to 6	12 varactors	2-D
[26]	2.39 to 2.47 (3%)	0° to -31°	8	Continuous	N/A	12 varactors	1-D
[28]	7.30	19° to 54°	10	Continuous	3.22 to 7.55	4 varactors	1-D
[29]	5.83 to 5.89 (1%)	0°	10	Discrete	5.8 to 8.3	2 coaxial ports	1-D
[38]	2.40 to 2.50 (4%)	0° to 90°	26	Discrete	N/A	4 capacitive coupling exciters	2-D
This work	5.00 to 7.00 (33.3%)	-40° to +40°	> 15	Continuous	4.6 to 7.78	4 varactors	1-D
	5.00 to 7.00 (33.3%)	-40° to +40°	> 10	Continuous	6.1 to 8.1	8 varactors	2-D

a small number of active elements to control the steering performance, namely four active varactors controlled by two bias voltages for the 1-D system, and eight varactors controlled by four bias voltages for the 2-D steering system and CP antenna system. Considering this performance, the proposed system outperforms others in literature.

In comparing various antenna systems across multiple parameters, the proposed work stands out in several key areas. The proposed system operates from 5 to 7 GHz, which is the broadest operating bandwidth among the listed references, covering an impressive 33.3% bandwidth. This is significantly wider than the ranges in [23] (2.21 to 2.59 GHz) and [24] (2.45 GHz), indicating superior flexibility in frequency handling. The null steering angular range extends from $-40°$ to $+40°$, which is the widest range observed among the compared studies, surpassing the ranges in [23] ($-28°$ to $28°$) and [24] ($0°$ to $32°$). The null depth achieved in this work is more than 15 dB for the 1-D configuration and 10 dB for the 2-D configuration, which is similar to the performance in [23] and [25]. In terms of gain, this work achieves a gain ranging from 4.6 to 8.1 dBi, which is higher than many other references, except [29], which achieves up to 8.3 dBi. Additionally, the proposed phase shifters employ four and eight varactors for reconfigurability, which, combined with its 2-D angular scanning capability, provides enhanced versatility over other references that only support 1-D scanning.

V. CONCLUSION

The design of a modular antenna system to provide versatile functionality in multidimensional null steering and CP generation has been presented. The system utilizes a compact, low-cost and high-performance tunable phase shifter, integrated with wideband printed dipoles. This integration facilitates both 1-D and 2-D null steering, as well as the generation of reconfigurable left-hand CP or right-hand CP. The compact phase shifter demonstrates notable performance with a phase tuning range of 145°, minimal phase imbalance of $\pm 2°$, and an insertion loss lower than 1 dB across a wide band extending from 5 to 7 GHz. The integration with a printed dipole array allows for effective 1-D null steering with a range of $\pm 40°$ and more than 15 dB null depth, while the 2-D system extended the null steering range to $\pm 40°$ in both *xz* and *yz* planes. Additionally, the system's capability to produce both left-hand CP and right-hand CP was achieved through precise phase adjustments, demonstrating high polarization purity. Experimental results confirmed the design's practicality, with measurements aligning well with simulations. This work highlights the potential of compact, tunable phase shifters in enhancing reconfigurable antenna systems, offering significant improvements in null steering, polarization control, and communications reliability.

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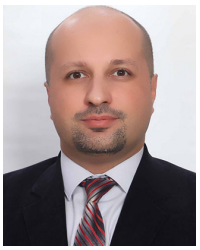
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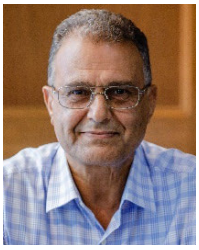
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Prof. Abbosh won many awards, for his excellence in fundamental, applied, and translational research, including IEEE APS King Prize (twice), the Excellence in HDR Supervision, and the Entrepreneurship Award.

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